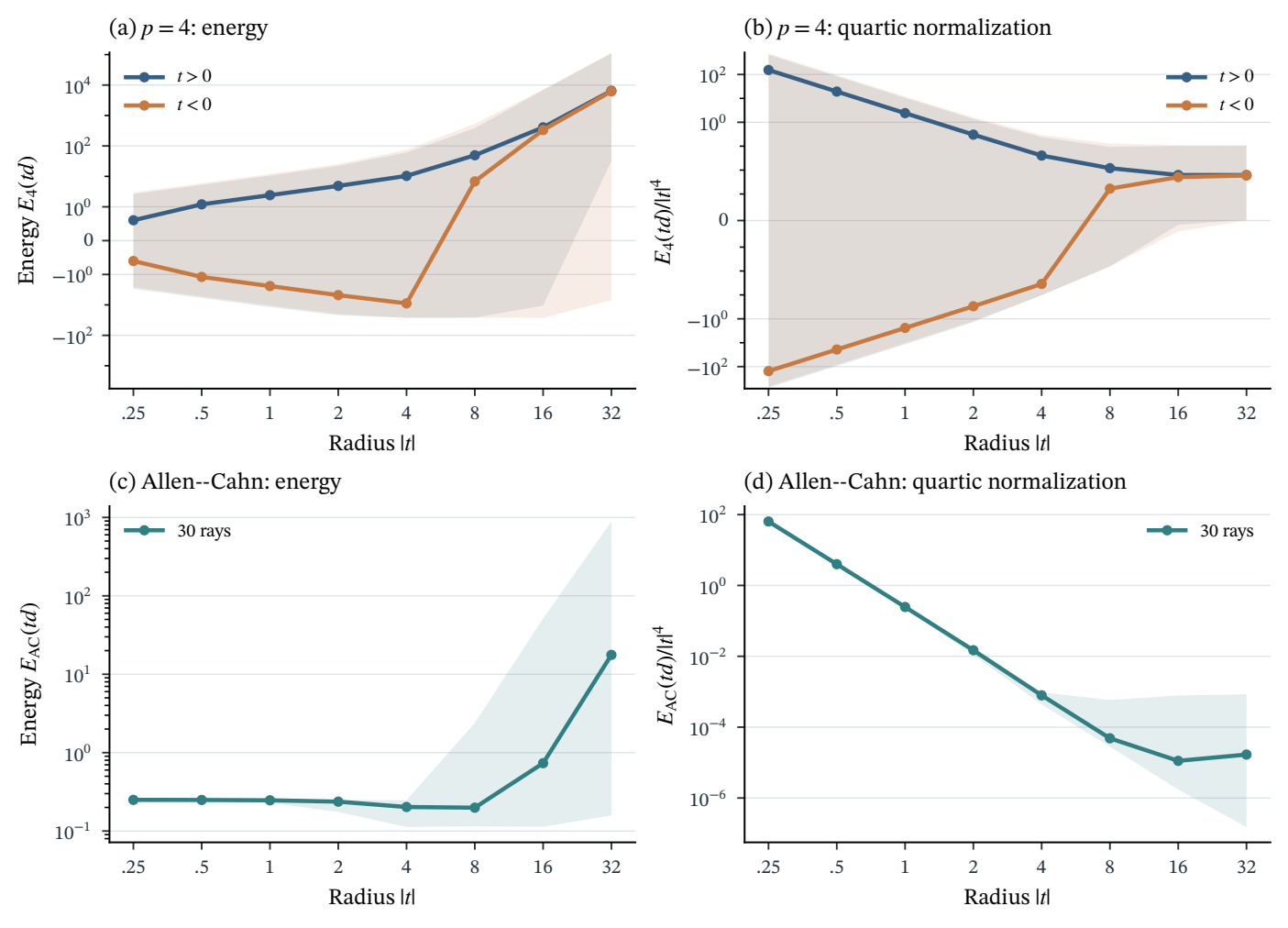
(a) $p = 4$: energy
$t > 0$
$t < 0$
Energy $E_4(td)$
10^4
10^2
10^0
0
-10^0
-10^2
.25
.5
1
2
4
8
16
32
Radius $|t|$
(b) $p = 4$: quartic normalization
$t > 0$
$t < 0$
$E_4(td)/|t|^4$
10^2
10^0
0
-10^0
-10^2
.25
.5
1
2
4
8
16
32
Radius $|t|$
(c) Allen--Cahn: energy
30 rays
Energy $E_{AC}(td)$
10^3
10^2
10^1
10^0
10^{-1}
.25
.5
1
2
4
8
16
32
Radius $|t|$
(d) Allen--Cahn: quartic normalization
30 rays
$E_{AC}(td)/|t|^4$
10^2
10^0
10^{-2}
10^{-4}
10^{-6}
.25
.5
1
2
4
8
16
32
Radius $|t|$